MyLearna
Plan • Capture • Grow

MATHEMATICS

Name: ____________________

Date: ____________________

Work with standard form and very large or very small numbers

Step 51

Use and interpret numbers written in standard form.

Today I am learning to work with standard form and very large or very small numbers.

1. Write in Standard Form

Write each number in standard form.

a) 56,000 = ____________

b) 7,800,000 = ____________

c) 0.0045 = ____________

d) 0.000072 = ____________

e) 123,000 ,000 = ____________

f) 0.00000089 = ____________

Think about it!

- Standard form looks like $a \times 10^n$ where $1 \le a < 10$ and n is an integer.
- Move the decimal to get one non-zero digit to the left.
- The number of places you move gives the power of 10.

2. Write in Standard Form (Scientific Notation)

Write each number in standard form.

a) 3,450,000,000 = ____________

b) 0.00000056 = ____________

c) 98,700,000 = ____________

d) 0.00045 = ____________

e) 6,020,000,000,000 = ____________

f) 0.00000000031 = ____________

Remember:

- Positive n for numbers ≥ 10
- Negative n for numbers < 1
- Count zeros carefully!

3. Convert Standard Form to Standard Form

Write each number in standard form.

a) 2.4×10^3 = ____________

b) 7.8×10^6 = ____________

c) 6.02×10^{-4} = ____________

d) 3.5×10^9 = ____________

e) 9.1×10^{-7} = ____________

f) 4.006×10^{12} = ____________

Think about it!

- If n is positive, move the decimal right n places.
- If n is negative, move the decimal left $|n|$ places.

4. Compare

Compare the numbers using <, >, or =.

a) 3.2×10^5 ______ 2.9×10^5

b) 7.5×10^{-3} ______ 2.1×10^{-2}

c) 6.7×10^8 ______ 6.7×10^6

d) 4.03×10^{-6} ______ 4.3×10^{-6}

e) 5×10^{10} ______ 9.9×10^9

f) 1.2×10^{-4} ______ 1.2×10^{-5}

Remember:

- Compare the powers of 10 first.
- If powers are the same, compare the leading numbers.

5. Order

Order the numbers from least to greatest.

a) 2.6×10^3, 2.6×10^5, 2.6×10^4, 2.6×10^2

______ < ______ < ______ < ______

b) 5.6×10^{-3}, 6.1×10^{-4}, 5.56×10^{-2}, 4.9×10^{-3}

______ < ______ < ______ < ______

c) 1.3×10^7, 1.3×10^6, 1.3×10^8, 1.3×10^5

______ < ______ < ______ < ______

d) 9.2×10^{-6}, 9.2×10^{-5}, 9.2×10^{-4}, 9.2×10^{-3}

______ < ______ < ______ < ______

Think about it!

- For positive powers, bigger power means bigger number.
- For negative powers, bigger power means smaller number.

6. Real-World Problems

Solve using standard form.

a) The distance from Earth to the Sun is about 149,600,000 km. Write this number in standard form.

Answer: ____________

b) A bacterium is about 0.0000025 m long. Write this number in standard form.

Answer: ____________

c) The mass of the Earth is about 5,972,000,000,000,000,000,000,000 kg. Write this number in standard form.

Answer: ____________

d) The diameter of a human hair is about 0.00008 m. Write this number in standard form.

Answer: ____________

Think about it!

- Standard form helps us work with numbers that are very large or very small.
- It makes calculations easier and comparisons clearer.

7. Use in Calculations

Calculate. Give your answer in standard form.

a) $(2.5 \times 10^3) \times (4 \times 10^2)$ = ____________

b) $(6.3 \times 10^6) \div (3 \times 10^2)$ = ____________

c) $(4.5 \times 10^{-4}) \times (2 \times 10^{-3})$ = ____________

d) $(9.6 \times 10^8) \div (1.2 \times 10^4)$ = ____________

Remember:

- When multiplying, multiply the numbers and add the powers.
- When dividing, divide the numbers and subtract the powers.

My Learning Check

Circle how confident you feel.

I can do it! | Almost there! | I need more practice. | I need help.

Parent Observation (optional)

- ☐ Understands the concept
- ☐ Uses standard form correctly
- ☐ Compares numbers accurately
- ☐ Solves problems correctly
- ☐ Explains thinking clearly
- ☐ Needs more practice

Notes: ____________

Today I learned...

1 thing I did well: ____________

1 thing I will improve: ____________